

PHARMACOLOGY ASSIGNMENT

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CLASS: BSN 4TH SEMESTER
SUBJECT: PHARMACOLOGY

1. CHOLINERGIC DRUGS (PARASYMPATHOMIMETICS)

Definition: Cholinergic drugs mimic the action of acetylcholine and stimulate the parasympathetic nervous system.

Classification:

- A. Direct Acting: Bethanechol, Pilocarpine, Carbachol, Methacholine.
- B. Indirect Acting (Anticholinesterases): Neostigmine, Physostigmine, Pyridostigmine, Donepezil.

Mechanism of Action:

Direct acting drugs stimulate muscarinic receptors directly. Indirect acting drugs inhibit acetylcholinesterase enzyme, increasing acetylcholine concentration at synapses.

Therapeutic Uses:

Urinary retention, glaucoma, myasthenia gravis, Alzheimer's disease.

Side Effects:

Salivation, Lacrimation, Urination, Diarrhea, GI cramps, Emesis, Miosis, Bradycardia.

Mnemonic: SLUDGE = Salivation, Lacrimation, Urination, Diarrhea, GI upset, Emesis.

2. ANTICHOLINERGIC DRUGS (PARASYMPATHOLYTICS)

Definition: Anticholinergic drugs block the action of acetylcholine at muscarinic receptors and inhibit parasympathetic activity.

Classification:

Atropine, Scopolamine, Ipratropium, Tiotropium, Tropicamide, Benztropine, Oxybutynin.

Mechanism of Action:

Block muscarinic receptors leading to reduced secretions, relaxation of smooth muscles and increased heart rate.

Uses:

Bradycardia, COPD, asthma, motion sickness, Parkinsonism, overactive bladder.

Side Effects:

Dry mouth, blurred vision, urinary retention, constipation, tachycardia, confusion.

Mnemonic: Can't See, Can't Pee, Can't Spit, Can't Shit.

3. ADRENERGIC DRUGS (SYMPATHOMIMETICS)

Definition: Adrenergic drugs stimulate alpha and beta adrenergic receptors and mimic sympathetic nervous system activity.

Classification:

Alpha Agonists: Phenylephrine, Clonidine.

Beta-1 Agonists: Dobutamine.

Beta-2 Agonists: Salbutamol, Terbutaline.

Mixed Agonists: Epinephrine, Norepinephrine, Dopamine.

Mechanism of Action:

Stimulate adrenergic receptors causing vasoconstriction, bronchodilation, increased cardiac output and increased blood pressure.

Uses:

Shock, cardiac arrest, asthma, hypotension, heart failure.

Side Effects:

Hypertension, tachycardia, palpitations, anxiety, tremors, insomnia.

Mnemonic: Fight or Flight Drugs.

4. ANTI-ADRENERGIC DRUGS (SYMPATHOLYTICS)

Definition: Anti-adrenergic drugs block sympathetic nervous system activity by inhibiting alpha or beta receptors.

Classification:

Alpha Blockers: Prazosin, Doxazosin, Phentolamine.

Beta Blockers: Propranolol, Atenolol, Metoprolol, Carvedilol.

Mechanism of Action:

Block adrenergic receptors causing reduced heart rate, blood pressure and myocardial oxygen demand.

Uses:

Hypertension, angina, arrhythmias, heart failure, benign prostatic hyperplasia.

Side Effects:

Bradycardia, hypotension, fatigue, dizziness, depression, bronchospasm.

Mnemonic: Beta Blockers = Slow the Beat.

COMPARISON TABLE (VIVA FAVORITE)

Cholinergic: Rest and Digest, decreases heart rate, increases secretions.

Anticholinergic: Opposite of cholinergic, dries secretions, increases heart rate.

Adrenergic: Fight and Flight, increases heart rate and blood pressure.

Anti-Adrenergic: Opposes sympathetic activity, lowers heart rate and blood pressure.

High Yield Exam Points:

- Atropine is the prototype anticholinergic drug.
- Neostigmine is used in myasthenia gravis.

- Salbutamol is a beta-2 agonist used in asthma.
- Propranolol is a non-selective beta blocker.
- Epinephrine is used in anaphylactic shock.
- Prazosin is an alpha-1 blocker used in hypertension.